

The Omega Block: correlation-network features that improve systemic tail-loss attribution beyond downside beta and conditional MES

Alberto Acedo. Biome Makers Inc.

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Abstract

Attributing future systemic tail losses across assets remains dominated by exposures estimated asset by asset, market beta, downside beta and volatility, although drawdowns are precisely the episodes in which losses become collective and concentrate where shocks recirculate among tightly co-moving names. The quantities proposed here originate in our prior work on network fragility: the FSRI, a structural index for microbial co-occurrence networks in agricultural soils, and its financial descendant, the TSI, whose standalone version we falsified and reported. From that starting point we hypothesized that the same structural information could carry marginal value if localized to individual assets, extended with a response dimension, and handed to a flexible learner as features rather than composed by hand. The result is the Omega Block: five per-asset features computed from daily prices alone, two triadic quantities measured against a closed-form configuration null and three response quantities around scheduled FOMC announcements, among them Omega-R, a perturbation response normalized by each asset's own baseline drift, developed for this block. In pre-registered paired ablations with a same-width noise placebo, the block adds 1.3 to 3.4 points of out-of-fold R^2 to next-year tail-loss attribution over baselines that instantiate the public state of the art, up to 31 columns including sector, downside beta, co-moments, a conditional MES proxy, size, liquidity and balance-sheet fundamentals, replicated on two universes and two nearly disjoint periods (2012-2017, 2016-2025), while the placebo worsens the same model in every configuration. Because the baselines that decide adoption in practice are proprietary, the paper's verification is designed to travel to the data: an open acceptance harness reruns the entire protocol on any universe, baseline and event calendar, with a values-only mode and a cryptographic time-lock. The scope established by the evidence is cross-sectional attribution, not market timing.

1. Introduction

Systemic risk measurement asks two questions: how much stress the system will experience, and who will carry it. The second, the cross-sectional attribution of future tail losses, underpins stress-testing narratives, watchlists and hedging priorities, and its standard answer ranks assets by market beta, downside beta or trailing volatility (Acharya et al., 2017; Adrian and Brunnermeier, 2016; Ang, Chen and Xing, 2006; Brownlees and Engle, 2017), even though tail risk itself is known to price in the cross-section (Kelly and Jiang, 2014). These exposures are estimated asset by asset, yet a drawdown is the regime in which behavior becomes collective: losses concentrate where shocks recirculate among tightly co-moving names rather than dispersing. An improvement in the who of tail attribution, even a modest one, propagates to the feature layer of risk platforms, which is where such models evolve in practice.

Financial correlation networks have been studied for decades, from hierarchical structure and dynamic asset trees in correlation matrices (Mantegna, 1999; Onnela et al., 2003) to econometric connectedness across institutions (Billio et al., 2012; Diebold and Yilmaz, 2014). The literature is rich in description and thin in marginal value: few network quantities have been shown to improve a decision-relevant task out of sample once the standard exposures are in the model, and fewer still against a control that prices in what a flexible learner extracts from any additional columns (Gu, Kelly and Xiu, 2020), a concern sharpened by the multiple-testing critique of the factor literature (Harvey, Liu and Zhu, 2016). Our own prior work reached the

same wall from the inside: an extended pre-registered falsification campaign on price-derived network indices, reported separately, found that static functionals of the correlation matrix are absorbed by volatility, strength and beta whenever they are used as indices, alone or in linear combinations.

The quantities tested here predate their financial use. The FSRI was developed to measure structural fragility in microbial co-occurrence networks in agricultural soils, where triadic structure and its deviation from a degree-preserving null discriminate management regimes (Ortiz-Alvarez et al., 2021; Saati-Santamaria et al., 2026); the same functional was later validated as a training regularizer for language models (Acedo, 2026a) and formulated for equity correlation networks as the TSI (Acedo, 2026b). The falsification of the TSI as a standalone index was the starting point of this work: it localized the failure precisely (whatever a global static functional summarizes, volatility and strength already carry) and it suggested the hypothesis that the same structural information might survive in three moves, localizing the functional to individual assets, adding a response dimension that no single-matrix functional can express, and letting a supervised learner compose the pieces against the outcome instead of fixing a formula. This paper tests that hypothesis in the scenarios where the prior failures were sharpest, United States large caps over two nearly disjoint decades, and carries over a habit acquired in a field where spurious network signals are endemic (Faust and Raes, 2012): written pre-registration of every decision rule, and a same-width noise placebo under every claimed gain.

The paper contributes the Omega Block, a set of five per-asset features computable from prices alone, including the Omega-R response construction developed for it; the evidence, a replicated, sector-robust, protocol-robust increment over baselines that instantiate the public state of the art; and a verification protocol designed for a field in which the relevant baselines are proprietary, released as an open acceptance harness with a values-only signal-trial mode and a cryptographic time-lock.

Taken together, the paper turns a falsified index into a validated feature block: five columns computed in minutes from data any risk platform already holds, a gain that survives every public benchmark we could construct and replicates across two decades at nearly identical point estimates, and a verification protocol that lets any reader re-derive the result on their own data in a day. Section 2 presents the construction, Section 3 the evidence, and Sections 4 to 6 the interpretation, limits, and licensing terms under which the block is available to risk platforms.

2. Construction of the Omega Block

2.1 Mathematical construction

Let r denote daily log returns of N assets. Over a trailing 250-day window, let A be the absolute Pearson correlation matrix with zero diagonal, $s = A1$ the strength vector, and S the sum of s . The Omega Block consists of five features per asset; Omega-R (Q3 to Q5 below) is the response construction inside the block, not a separate object.

The two static quantities localize the triadic functional $\text{Tr}(A^3)$ of the Omega-S family (Acedo, 2026a; Acedo, 2026b) to individual assets. The triadic concentration of asset i is

$$tri_i = (A^3)_{ii} \quad (1)$$

the weighted count of closed triangles through i . Its excess over a configuration null in the sense of Newman (2006) and of the expected-degree model of Chung and Lu (2002), preserving all strengths in closed form,

$$E_i = s_i^2 (\sum_j s_j^2)^2 / S^3 \quad (2)$$

$$tri-exc_i = (tri_i - E_i) / E_i \quad (3)$$

isolates structure beyond what the asset's own connectedness explains.

The three response quantities compare each asset's position in the graph before and after scheduled macroeconomic announcements. For an announcement at day t , the rewiring response of asset i is the cosine distance between its correlation profiles over the 20 trading days before and after the event,

$$resp_i(t) = 1 - \cos(row_i(A_{pre}), row_i(A_{post})) \quad (4)$$

with A_{pre} estimated on $[t-21, t-2]$, A_{post} on $[t+1, t+20]$, diagonal excluded. Q3 is the median of (4) over the scheduled FOMC decision days inside the estimation window. Q4, the baseline drift, is the same median over a pool of placebo days at least 10 trading days from any announcement, drawn five times the number of events; the enlarged pool stabilizes per-asset values against the draw (Section 4). Q5, the normalized response we call Omega-R, is

$$omr_i = Q3_i / Q4_i \quad (5)$$

the perturbation response normalized by the asset's own rewiring rate absent identified shocks. Values near 1 indicate that announcements reorganize the asset's neighborhood no more than an ordinary fortnight does; values far from 1 indicate an event-sensitive position in the graph. Because (4) is a functional of two matrices, Omega-R sits outside the equivalence arguments that make single-matrix functionals redundant with spectral summaries; empirically its rank correlation with beta, volatility, strength and past MES stays below 0.4.

The composition of the five features is learned, not fixed. Hand-fixed compositions of these quantities were tested extensively in the falsification campaign that preceded this work and did not survive; shipping the block as coordinates and letting a supervised learner compose them out of fold is what the evidence supports.

2.2 Data

Universe A comprises 465 S&P constituents with at least 95 percent coverage of daily closes from December 2015 to December 19, 2025 (2,514 trading days), retrieved from Yahoo Finance. Universe B comprises 459 constituents from August 2012 to August 2017 (1,257 days), from the public mirror of Sidi (2017). Returns are log differences of daily closes. Sector labels are current GICS sectors; in Universe B, 140 delisted or renamed tickers group as Other, a declared limitation. The event calendar contains the 78 (A) and 39 (B) scheduled FOMC decision days, whose market-wide relevance is well documented (Lucca and Moench, 2015), hand-coded from the Federal Reserve's public record; the March 2020 emergency meetings are excluded because their timing is endogenous to the crash. Balance-sheet fundamentals for the control of Section 3.4 come from the SimFin bulk dataset (SimFin, 2026), using only statements published before each window's close. The target is each asset's average return on the market's worst 5 percent of days over the following 250 days, its realized future marginal expected shortfall (MES) against the equal-weight market (Acharya et al., 2017).

2.3 Validation protocol

The model is a HistGradientBoostingRegressor (300 trees, depth 3, learning rate 0.05), identical in every arm and never tuned, following the boosted-trees-over-characteristics paradigm of Gu, Kelly and Xiu (2020). Windows are 250 days in and 250 out, overlapping at a 21-day step for inference (96 windows in A, 37 in B) and non-overlapping for consistency counts (9 and 4). Evaluation uses 5 folds across assets inside each window, so the model never sees the test assets; both arms share folds and period, and the statistic is the paired difference in out-of-fold R^2 when a block is added to the baseline. Every ablation is accompanied by a placebo arm that adds five Gaussian noise columns under the identical protocol: with boosting, width without signal is harmful, so the placebo provides a negative floor that any genuine gain must clear by a visible margin. Inference is a paired block bootstrap over overlapping windows (blocks of 12, 2,000 resamples). Each

experiment was pre-registered with a written decision rule before running; the files ship with the code. Figure 1 summarizes the pipeline.

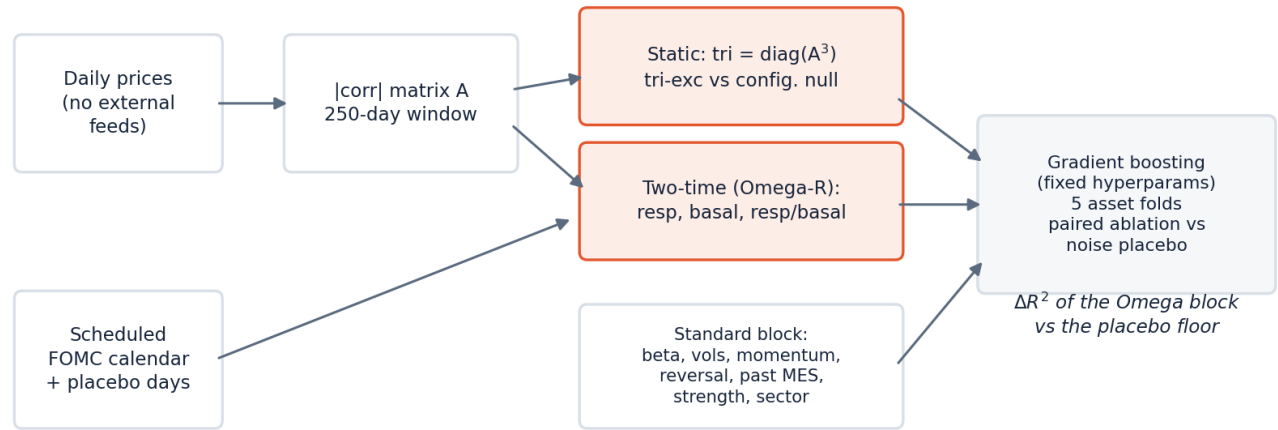


Figure 1. The Omega Block pipeline: from prices and the announcement calendar to the placebo-gated ablation.

2.4 Benchmark feature sets

All baselines are feature sets computed from the same price panel; they differ in which published characteristics they include, not in data source. The standard block (7 columns) contains market beta, 21- and 250-day volatility, momentum, reversal, past MES and strength. The sector-controlled set adds 11 GICS dummies and Other. The maximal public set (24) instantiates the published state of the art in price-computable tail characteristics: downside beta (Ang, Chen and Xing, 2006), coskewness and cokurtosis (Harvey and Siddique, 2000), downside semideviation, and an EWMA (lambda 0.94) conditional MES forecast in the spirit of the V-Lab methodology (Brownlees and Engle, 2017). The size and liquidity control (26) adds log average dollar volume and log Amihud (2002) illiquidity. The fundamentals control (31, sub-period) adds book leverage, quasi-market leverage, log assets, book-to-market and trailing profitability from published quarterly balance sheets. Improvement over these sets is therefore improvement over the existing public methods for the task.

3. Results

3.1 The block improves attribution and the gain replicates across decades

On the standard block the gain is +0.034 (CI95 +0.027 to +0.042) in Universe A and +0.021 (+0.010 to +0.033) in Universe B, positive in 8 of 9 non-overlapping windows in A. The placebo arm is negative in every configuration and both universes, between -0.005 and -0.018, so the observed gains sit 2 to 5 points above what the addition of columns per se produces. Table 1 and Figure 2 report all configurations.

baseline	universe	Omega block	placebo block
standard (7)	A: 465, 2016-25	+0.034 (+0.027, +0.042)	-0.016 (-0.020, -0.012)
standard (7)	B: 459, 2012-17	+0.021 (+0.010, +0.033)	run with sector below
standard + sector	A	+0.017 (+0.011, +0.022)	-0.018 (-0.025, -0.012)
standard + sector	B	+0.016 (+0.005, +0.028)	-0.008 (-0.013, -0.001)
maximal public (24)	A	+0.015 (+0.010, +0.020)	-0.015 (-0.023, -0.009)
maximal public (24)	B	+0.015 (+0.003, +0.027)	-0.012 (-0.021, -0.004)
+ size/liquidity (26)	A	+0.013 (+0.008, +0.018)	-0.012 (-0.018, -0.007)

baseline	universe	Omega block	placebo block
+ size/liquidity (26)	B	+0.013 (+0.006, +0.021)	-0.005 (-0.009, +0.001)
+ fundamentals (31)	A, 2021-25 sub	+0.013 (+0.001, +0.025)	-0.006 (-0.008, -0.003)

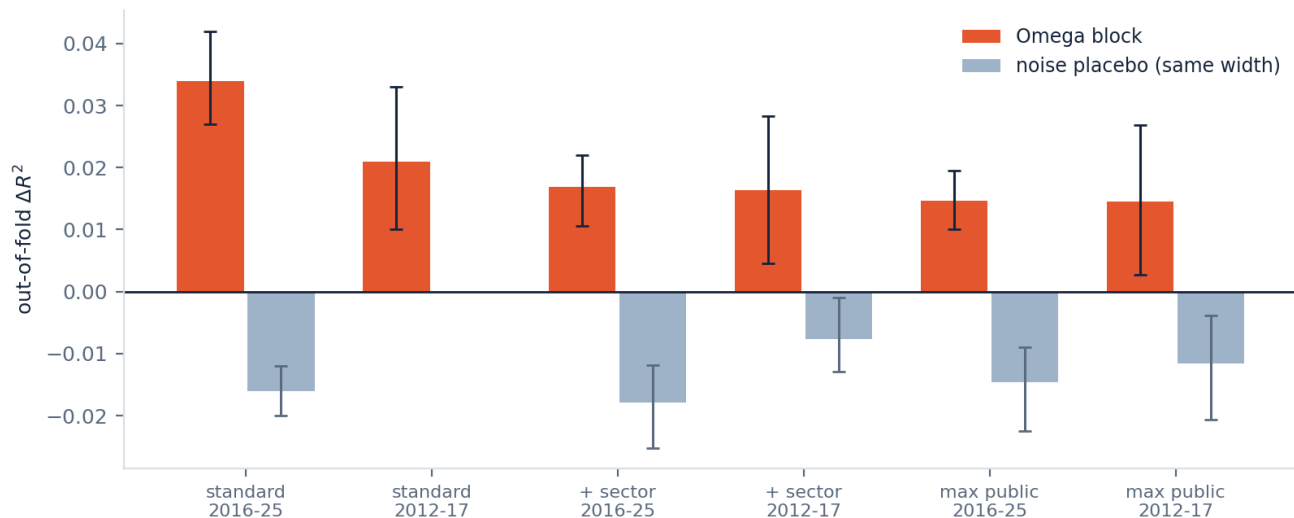


Figure 2. Omega Block versus the noise placebo across baselines and universes, with bootstrap CI95. The placebo below zero is the floor every gain must clear.

3.2 Sector membership absorbs half of the raw gain, and the half that survives is stable

Adding GICS dummies quantifies how much of the naive effect was sector membership: half. The surviving half, +0.017 in A and +0.016 in B, agrees across two nearly disjoint decades almost to the third decimal.

3.3 The block beats the public state of the art in tail characteristics

Against the 24-column set that instantiates the published methods, downside beta, co-moments, semideviation and the conditional MES proxy, the block still contributes +0.015 in both universes with confidence intervals excluding zero. Because index membership and trading activity could plausibly explain triadic density, the next control adds size and liquidity: the block survives with an identical +0.013 point estimate in both universes. No public tail-risk characteristic computable from prices carries the information in the block.

3.4 Leverage and fundamentals do not absorb the effect

On the 2021-2025 sub-period where published quarterly balance sheets cover roughly 300 constituents, adding book and market leverage, log assets, book-to-market and trailing profitability leaves the estimate unchanged (+0.013 against +0.012 for the same windows without fundamentals). The remaining fundamentals frontier, the full period and Universe B, is untested for lack of free coverage and is stated as such.

3.5 The gain has an interpretable source

A decomposition declared post hoc explains where the result comes from. The static pair (Q1, Q2) carries most of the gain, +0.030 (+0.024, +0.034); the Omega-R response set adds +0.009 (+0.003, +0.018); the two are approximately additive and both survive excluding the covid windows. Q1 alone is nearly collinear with strength, which the baseline already contains: its contribution materializes only jointly with the null-adjusted Q2 and only through interactions, which is why an extensive linear literature could not have found it, and which is consistent with dense triadic neighborhoods being the configurations in which shocks recirculate rather than disperse. Robustness was assessed over three configurations fixed in advance (depth 2 with learning rate 0.10, +0.043; depth 4 with 200 trees, +0.033; a different fold seed, +0.036), positive in 7 to 8 of 9 independent windows in every configuration: the result does not depend on a fortunate corner of hyperparameter space.

3.6 The gain translates into decisions

Averaged over the nine independent windows of Universe A, the base model places 20.9 of the true 50 heaviest next-year stress carriers in its predicted top 50; with the block the count rises to 22.8, two additional true carriers identified per year, improving in six of nine windows (chance would place 5.4). Out-of-fold rank correlation with realized tail losses rises from 0.674 to 0.705. In the last completed cycle (features through November 2024, outcomes through December 2025) the out-of-fold R^2 rises from 0.601 to 0.632, and the per-asset marginal attributable to the block spans roughly plus or minus 8 basis points of daily tail loss, concentrated in names whose triadic excess departs most from the null. Figure 3 summarizes both views.

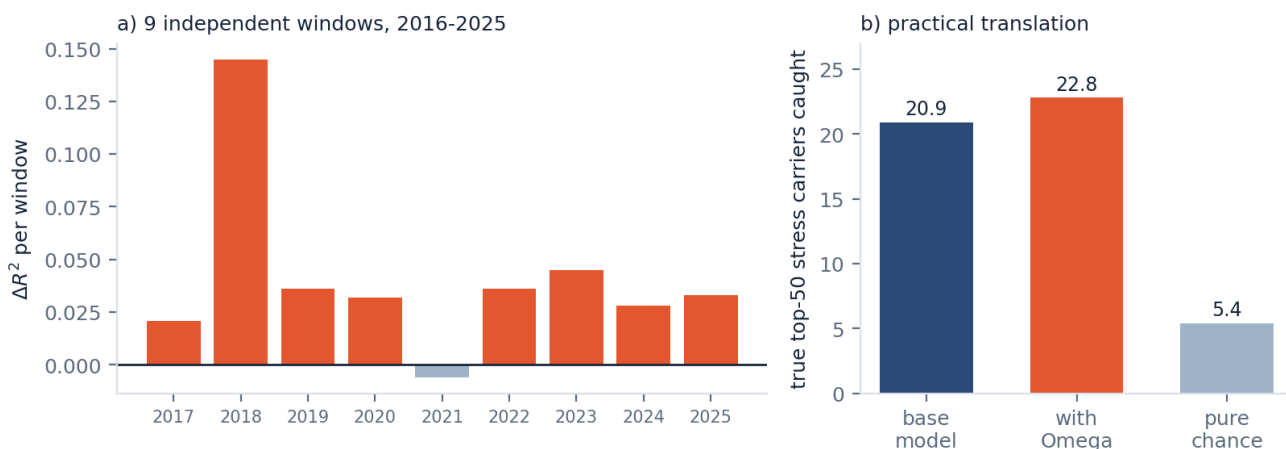


Figure 3. a) Per-window increment across the nine independent windows. b) The gain translated to decisions: true top-50 stress carriers caught.

3.7 The forward route is closed by the evidence

Two forward designs, training strictly on past windows, were pre-registered, the second declared as a second attempt with its multiplicity acknowledged. In raw levels the baseline itself fails to transfer across regimes (out-of-time R^2 between -4.5 and +0.5), rendering the first test uninformative about the block. With within-window rank standardization the baseline transfers well (R^2 +0.30 to +0.63) and the block adds nothing forward (3 of 8 positive, mean -0.013, Wilcoxon $p = 0.68$). The scope of the paper is therefore cross-sectional attribution: who carries stress, not when it arrives.

3.8 The verification protocol is itself a result

The baselines that decide adoption are proprietary: every risk platform maintains factor sets and estimators that outside researchers cannot access, so no fixed set of published tables, ours included, can settle the question that matters to a practitioner, whether the block adds value on their universe against their model. We treat this data limitation as a design constraint and make the verification itself portable (Figure 4). The released harness reruns the complete protocol, folds, placebo floor, block bootstrap and acceptance rule, on any wide CSV of prices, any baseline columns and any event calendar, and prints an explicit pass or fail. A values-only signal trial allows the converse direction: an evaluator receives the block as a plain numbers file and tests it without any formula or code of ours entering their environment; we verified this mode blind, reproducing the headline +0.036 against the -0.016 placebo. For look-ahead skepticism, future feature values can be committed by publishing the SHA-256 hash of the coming quarter's file, timestamped, and revealing the file once outcomes are realized. Code, harness, pre-registrations and the replication registry are available at github.com/BiomeMakers/Omega-block.

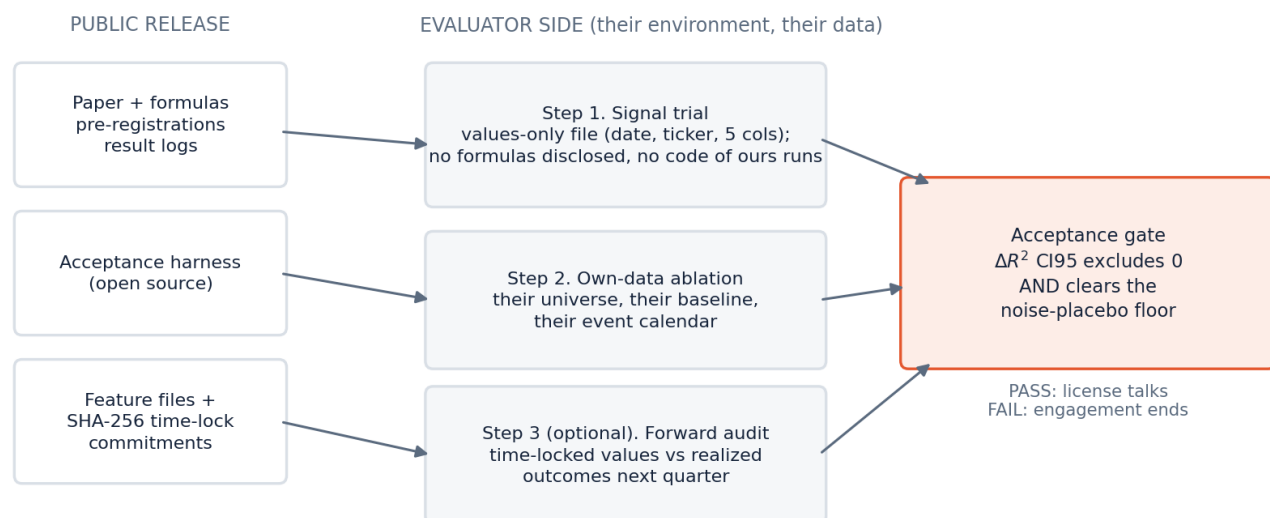


Figure 4. The verification process and its pieces: public release, values-only signal trial, own-data ablation, optional time-locked forward audit, and the acceptance gate.

4. Discussion

The mechanism the block appears to capture is recirculation. An asset whose triadic neighborhood is denser than its strength explains sits where shocks bounce among closely co-moving names instead of dispersing, and the boosted model exploits that property only in interaction with the standard exposures, which is consistent with the absence of the effect in every linear test we and the literature have run, and with network models of contagion in which dense interconnection turns from shock-absorbing to shock-propagating (Acemoglu, Ozdaglar and Tahbaz-Salehi, 2015). Omega-R contributes an orthogonal axis, distinguishing assets whose correlation neighborhood absorbs scheduled announcements from assets whose neighborhood reorganizes; its rank correlation below 0.4 with every standard exposure is, to our knowledge, unique among published network quantities on price data.

For practice, the block targets the layer where risk platforms actually improve, the feature set of a model release. Integration requires no new data licenses, because every input is computed from daily closes the platform already holds plus a public announcement calendar, and no meaningful compute. The immediate applications follow from the task: sharper systemic-stress watchlists, per-name attribution to enrich stress narratives presented internally and to supervisors, and a structure-sensitive complement to downside beta in tail-risk screens.

The conclusions the evidence supports are three. The Omega Block improves the cross-sectional attribution of future tail losses over every public benchmark we could construct, from the standard exposures to a 31-column replica of a vendor feature set, with a negative placebo floor under every comparison and point estimates that replicate across two nearly disjoint decades. The improvement is modest by construction, two additional true top-50 stress carriers per year on a 465-name book, and we frame it as a feature-level contribution: on public price data, effects large enough to anchor a standalone product should not survive, and claims that they do deserve suspicion. And the definitive comparison, against the proprietary baselines of each platform, cannot be run by us, which is why this paper ships the application that runs it: the acceptance harness of Section 3.8 exists because the datasets that settle the question are private, and it converts that limitation into a test any interested party can execute on their own data in a day.

The limitations are stated where they bind. Both universes are US large caps; breadth across asset classes and geographies is untested. The event calendar is hand-coded, and dating errors bias toward the null. Sector

labels in Universe B are anachronistic for delisted names. The conditional MES benchmark is an EWMA proxy rather than a maximum-likelihood DCC fit; the harness accepts the full estimator as a drop-in replacement. Correlation matrices at these dimensions carry estimation noise; shrinkage estimators (Ledoit and Wolf, 2004) are the standard remedy, and the harness accepts shrunk inputs. The fundamentals control covers 2021-2025 and the covered sub-universe only. Per-asset values of Q3-Q5 are sensitive to the placebo draw when the pool is small (cross-seed rank correlation 0.52 at a pool equal to the event count); enlarging the pool five-fold raises it to 0.88 with the aggregate contribution stable under every draw tested, and the five-fold pool is therefore part of the definition of Q4. Finally, the evaluation shares the outcome period between folds, which is appropriate for attribution but must not be read as market timing; the forward tests of Section 3.7 keep that boundary honest. Within these limits, the invitation stands: in a field of proprietary baselines a result that cannot be re-derived on the reader's own data is only half a result, and the harness exists so that the other half can be supplied by anyone.

5. Intellectual property

The quantities Q1 to Q5 and their use as inputs to risk-attribution models are covered by the Omega-S patent family (USPTO provisionals 64/112,912 and 64/121,656, and the use-focused provisional 64/140,524, filed August 25, 2026). The formulas are disclosed in full in Section 2; the license offered to analytics platforms covers freedom to operate, the acceptance harness and the update stream of the family, not the disclosed mathematics as such.

6. Use of AI assistance

This preprint was prepared with the assistance of an AI system (Claude, Anthropic) working under the direction of the author, who designed the research program, set every decision rule and takes full responsibility for the content. The same disclosure practice is used across the author's preprint series.

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- [Pending: add the companion falsification-campaign paper when posted.]